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Status Finished

Started Tuesday, 6 October 2026, 9:21 AM

Completed Tuesday, 6 October 2026, 9:27 AM

Duration 6 mins 19 secs

Marks 16.00/16.00

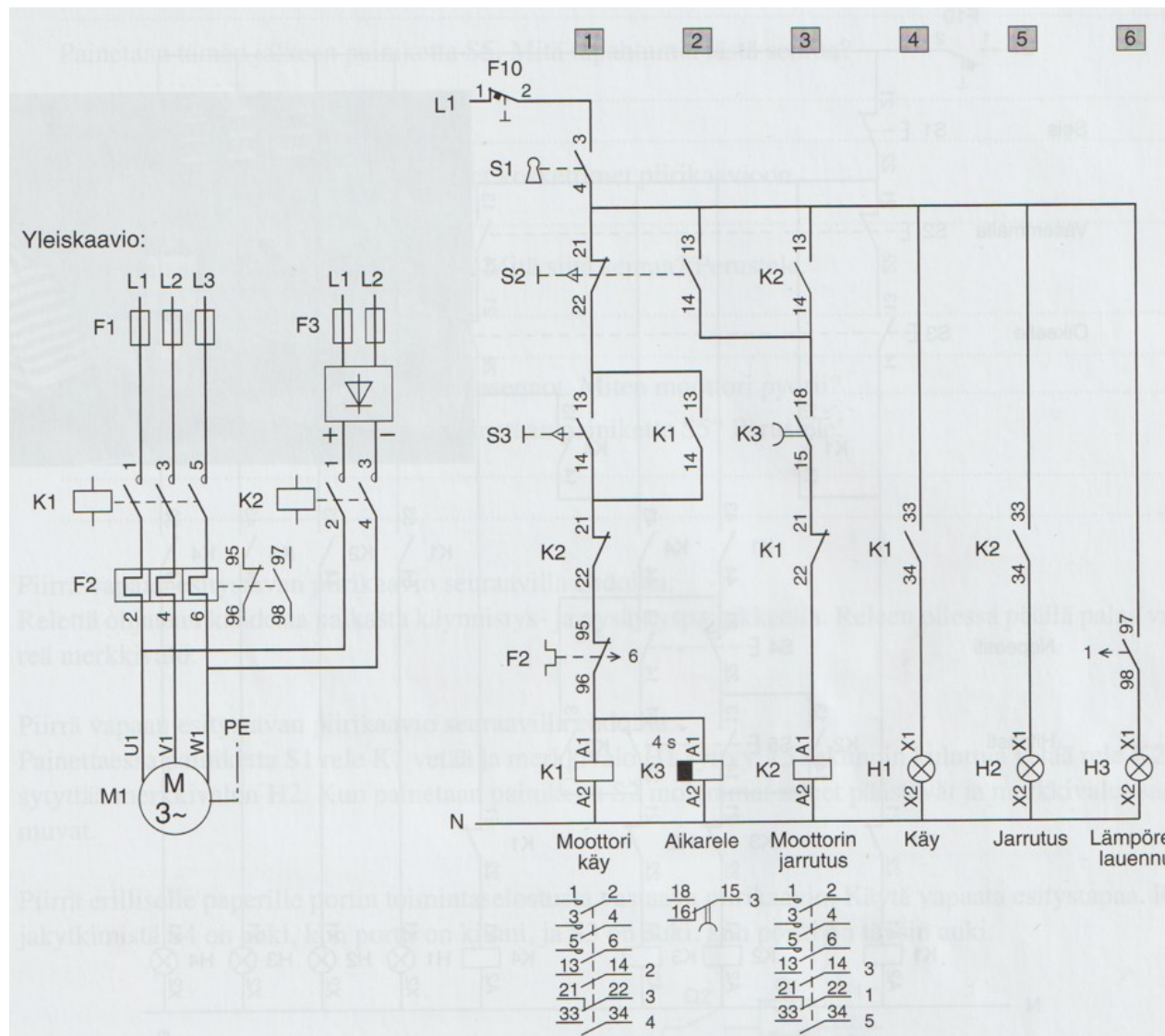
Grade 10.00 out of 10.00 (100%)

Question 1

Correct

Mark 4.00 out of 4.00

Complete the operation of the circuit shown below. The circuit diagram represents the DC braking of a three-phase **induction motor**. (Source: J. Ahoranta, Sähköasennustekniikka 2015)



Yleiskaavio → General Diagram

Moottori käy → Motor Running

Aikarele → Time Relay

Moottorin jarrutus → Motor Braking

Käy → Running

Jarrutus → Braking

Lämpörelä lauennut → Thermal Overload Tripped

Motor starting:

The key switch S1 is in the closed position. When pushbutton S3 is pressed, the coil of contactor **K1** is energized, and the contactor is held through its auxiliary holding contact **13-14**. In the power circuit, the main contacts 1, 3 and 5 of the contactor close, and the three-phase motor **M1** starts. Indicator lamp H1 (Running) is energized when contacts **33-34** of contactor K1 close.

Question 2

Correct

Mark 11.00 out of 11.00

Motor stopping:

To stop the motor, pushbutton **S2** ✓ is pressed. Contactor K1 **de-energizes** ✓, because 0-pushbutton S2 interrupts the contactor control circuit. Contact 33-34 of contactor K1 switches off indicator lamp **H1** ✓, and contact 21-22 energizes contactor **K2** ✓, because pressing S2 creates a closed circuit through S1, contact **13-14** ✓ of S2, contact **15-18** ✓ of K3, and contact **21-22** ✓ of K1.

Relay K2 receives a holding circuit through its auxiliary contact 13-14 and, through contact **33-34** ✓, switches on indicator lamp **H2** ✓. At the same time, contact 21-22 interrupts the control circuit of contactor K1.

After a delay of 4 seconds, the **off-delay time relay** **K3** ✓ releases and opens its contact 15-18. As a result, contactor K2 de-energizes, indicator lamp H2 turns off, and motor braking ends.

The motor can then be started again using pushbutton **S3** ✓.

Question 3

Correct

Mark 1.00 out of 1.00

Why can the motor not be started again during the braking period?

- ☐ a. There is not enough current available to start the motor during braking.
- ☐ b. The open contacts 95-96 of the thermal overload relay prevent contactor K1 from energizing.
- ☒ c. The open contacts 21-22 of contactor K2 prevent contactor K1 from energizing. ✓
- ☐ d. Indicator lamp H2 is energized and prevents the motor from starting.